



Swiggy vs Zomato: What Drives Restaurant Success in India?

Exploratory Data Analysis on Customer Experience, Profitability & Delivery Performance

Notebook Description

This notebook presents a comprehensive Exploratory Data Analysis (EDA) of restaurant performance across India's two leading food delivery platforms — Swiggy and Zomato.

The analysis focuses on uncovering the key factors that influence restaurant success by examining financial performance, customer satisfaction, operational efficiency, delivery experience, pricing strategy, and platform competitiveness.

Through data-driven exploration and business-oriented insights, this notebook aims to answer questions such as:

- What drives higher restaurant profitability?
 - How does delivery performance impact customer ratings?
 - Which cities and cuisines perform best?
 - Are discounts helping growth or hurting margins?
 - Which platform demonstrates stronger operational and customer performance?
-



What To Expect From This Notebook

- Data Cleaning & Preparation
 - Univariate, Bivariate & Multivariate Analysis
 - Revenue & Profitability Insights
 - Customer Behavior Analysis
 - Delivery & Operational Performance Evaluation
 - Swiggy vs Zomato Competitive Comparison
 - Correlation & Trend Analysis
 - Business Recommendations & Strategic Insights
-

Intended Audience

This notebook is designed for:

- Data Analysts & BI Enthusiasts
 - Recruiters evaluating analytical projects
 - Business & Operations Professionals
 - Students learning EDA and storytelling
 - Anyone interested in food delivery market intelligence and platform analytics
-

Goal of the Analysis

The primary objective of this analysis is not just to visualize data, but to understand the business dynamics behind restaurant success in the online food delivery ecosystem and transform raw data into actionable insights.

```
In [2]: # 📖 Importing Required Libraries
```

```
import pandas as pd
import numpy as np
```

```
In [3]: # Visualization Libraries
```

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [4]: # 📁 Loading the Dataset
```

```
df = pd.read_csv("swiggy_vs_zomato_3000.csv")
```

```
In [5]: # Preview Dataset
```

```
df.head()
```

Out[5]:

	restaurant_id	restaurant_name	city	locality	restaurant_type
0	REST000001	Restaurant_1	Mumbai	Mumbai_Locality_9	Cafe
1	REST000002	Restaurant_2	Mumbai	Mumbai_Locality_13	Casual Dining
2	REST000003	Restaurant_3	Mumbai	Mumbai_Locality_12	Fine Dine
3	REST000004	Restaurant_4	Mumbai	Mumbai_Locality_19	Cloud Kitchen
4	REST000005	Restaurant_5	Mumbai	Mumbai_Locality_13	Cafe

5 rows × 41 columns

In [6]: `df.tail()`

Out[6]:	restaurant_id	restaurant_name	city	locality	restaurant_t
2495	REST002496	Restaurant_2496	Lucknow	Lucknow_Locality_10	Food Ti
2496	REST002497	Restaurant_2497	Lucknow	Lucknow_Locality_14	Fine I
2497	REST002498	Restaurant_2498	Lucknow	Lucknow_Locality_16	Food Ti
2498	REST002499	Restaurant_2499	Lucknow	Lucknow_Locality_9	(
2499	REST002500	Restaurant_2500	Lucknow	Lucknow_Locality_16	QSR (Q Serv

5 rows × 41 columns

Dataset Overview

Before diving into analysis, it is important to understand the structure, dimensions, and overall quality of the dataset. This section explores the dataset shape, column information, datatypes, and initial observations to build a strong foundation for further analysis.

In [7]: `df.describe()`

Out[7]:	distance_from_city_center_km	days_operational	swiggy_rating	swiggy_t
count	2500.00000	2500.000000	2500.000000	
mean	7.53788	1844.268400	3.985640	
std	4.23873	1044.579864	0.400917	
min	0.50000	33.000000	2.500000	
25%	3.80000	946.250000	3.700000	
50%	7.40000	1846.000000	4.000000	
75%	11.20000	2728.250000	4.300000	
max	15.00000	3649.000000	5.000000	

8 rows × 26 columns

```
In [8]: # Shape of Dataset
print("Dataset Shape:", df.shape)

# Column Information
df.info()
```

```
Dataset Shape: (2500, 41)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2500 entries, 0 to 2499
Data columns (total 41 columns):
```

#	Column	Non-Null Count	Dtype
0	restaurant_id	2500 non-null	object
1	restaurant_name	2500 non-null	object
2	city	2500 non-null	object
3	locality	2500 non-null	object
4	restaurant_type	2500 non-null	object
5	cuisines	2500 non-null	object
6	distance_from_city_center_km	2500 non-null	float64
7	opening_time	2500 non-null	object
8	closing_time	2500 non-null	object
9	days_operational	2500 non-null	int64
10	swiggy_rating	2500 non-null	float64
11	swiggy_total_reviews	2500 non-null	int64
12	zomato_rating	2500 non-null	float64
13	zomato_total_reviews	2500 non-null	int64
14	average_rating_both_platforms	2500 non-null	float64
15	avg_cost_per_person_inr	2500 non-null	int64
16	price_category	2500 non-null	object
17	swiggy_delivery_fee_inr	2500 non-null	int64
18	zomato_delivery_fee_inr	2500 non-null	int64
19	swiggy_avg_delivery_time_minutes	2500 non-null	int64
20	zomato_avg_delivery_time_minutes	2500 non-null	int64
21	swiggy_platform_commission_pct	2500 non-null	float64
22	zomato_platform_commission_pct	2500 non-null	float64
23	swiggy_discount_frequency_pct	2500 non-null	int64
24	zomato_discount_frequency_pct	2500 non-null	int64
25	swiggy_estimated_monthly_orders	2500 non-null	int64
26	zomato_estimated_monthly_orders	2500 non-null	int64
27	swiggy_estimated_monthly_revenue_inr	2500 non-null	int64
28	zomato_estimated_monthly_revenue_inr	2500 non-null	int64
29	swiggy_estimated_net_profit_inr	2500 non-null	int64
30	zomato_estimated_net_profit_inr	2500 non-null	int64
31	swiggy_market_share_pct	2500 non-null	float64
32	zomato_market_share_pct	2500 non-null	float64
33	platform_performance_better	2500 non-null	object
34	amenities	2500 non-null	object
35	amenities_count	2500 non-null	int64
36	has_own_website	2500 non-null	bool
37	has_own_app	2500 non-null	bool
38	food_license_verified	2500 non-null	bool
39	listing_date	2500 non-null	object
40	days_listed	2500 non-null	int64

```
dtypes: bool(3), float64(8), int64(18), object(12)
```

```
memory usage: 749.6+ KB
```

```
In [9]: # Checking Missing Values
df.isnull().sum()
```

```

Out[9]: restaurant_id      0
        restaurant_name    0
        city               0
        locality           0
        restaurant_type    0
        cuisines            0
        distance_from_city_center_km  0
        opening_time       0
        closing_time       0
        days_operational   0
        swiggy_rating      0
        swiggy_total_reviews  0
        zomato_rating      0
        zomato_total_reviews  0
        average_rating_both_platforms  0
        avg_cost_per_person_inr  0
        price_category     0
        swiggy_delivery_fee_inr  0
        zomato_delivery_fee_inr  0
        swiggy_avg_delivery_time_minutes  0
        zomato_avg_delivery_time_minutes  0
        swiggy_platform_commission_pct  0
        zomato_platform_commission_pct  0
        swiggy_discount_frequency_pct  0
        zomato_discount_frequency_pct  0
        swiggy_estimated_monthly_orders  0
        zomato_estimated_monthly_orders  0
        swiggy_estimated_monthly_revenue_inr  0
        zomato_estimated_monthly_revenue_inr  0
        swiggy_estimated_net_profit_inr  0
        zomato_estimated_net_profit_inr  0
        swiggy_market_share_pct  0
        zomato_market_share_pct  0
        platform_performance_better  0
        amenities          0
        amenities_count    0
        has_own_website    0
        has_own_app        0
        food_license_verified  0
        listing_date       0
        days_listed        0
        dtype: int64

```



Data Cleaning & Validation

Before performing exploratory analysis, it is important to validate the dataset for duplicates, datatype consistency, and overall data quality. This step ensures that the analysis is reliable, accurate, and free from structural issues.

```

In [10]: # Checking Duplicate Rows
df.duplicated().sum()

```

```
Out[10]: np.int64(0)
```

```
In [11]: df.dtypes
```

```
Out[11]: restaurant_id      object
restaurant_name      object
city                  object
locality              object
restaurant_type       object
cuisines              object
distance_from_city_center_km  float64
opening_time          object
closing_time          object
days_operational      int64
swiggy_rating          float64
swiggy_total_reviews    int64
zomato_rating          float64
zomato_total_reviews    int64
average_rating_both_platforms  float64
avg_cost_per_person_inr  int64
price_category         object
swiggy_delivery_fee_inr  int64
zomato_delivery_fee_inr  int64
swiggy_avg_delivery_time_minutes  int64
zomato_avg_delivery_time_minutes  int64
swiggy_platform_commission_pct  float64
zomato_platform_commission_pct  float64
swiggy_discount_frequency_pct  int64
zomato_discount_frequency_pct  int64
swiggy_estimated_monthly_orders  int64
zomato_estimated_monthly_orders  int64
swiggy_estimated_monthly_revenue_inr  int64
zomato_estimated_monthly_revenue_inr  int64
swiggy_estimated_net_profit_inr  int64
zomato_estimated_net_profit_inr  int64
swiggy_market_share_pct  float64
zomato_market_share_pct  float64
platform_performance_better  object
amenities              object
amenities_count         int64
has_own_website          bool
has_own_app              bool
food_license_verified    bool
listing_date             object
days_listed             int64
dtype: object
```

```
In [12]: # Converting listing_date to datetime
df['listing_date'] = pd.to_datetime(df['listing_date'])
```

```
In [13]: df['listing_date'].dtype
```

```
Out[13]: dtype('<M8[ns]')
```



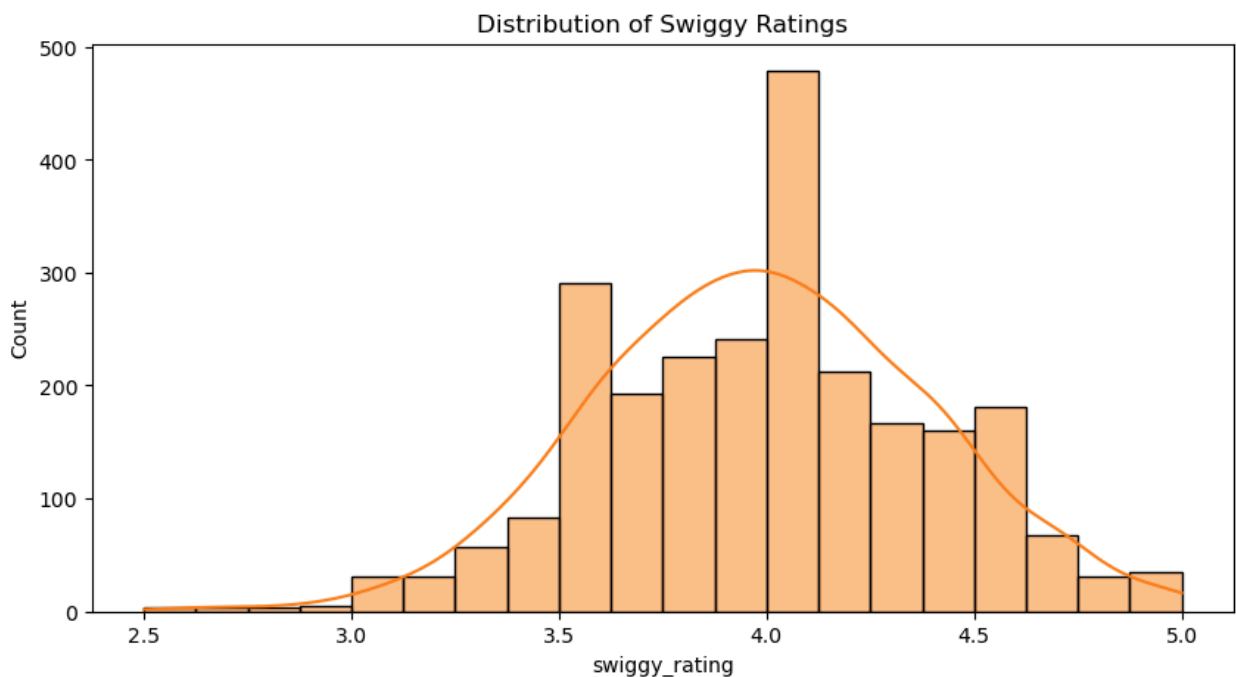

Univariate Analysis

Univariate analysis focuses on understanding the distribution and behavior of individual variables within the dataset. This section helps identify patterns, trends, skewness, and potential anomalies across financial, customer, operational, and competitive metrics.

```
In [14]: plt.figure(figsize=(10,5))

sns.histplot(
    df['swiggy_rating'],
    bins=20,
    kde=True,
    color='#FC8019',
    edgecolor='black'
)

plt.title("Distribution of Swiggy Ratings")
plt.show()
```



Observation

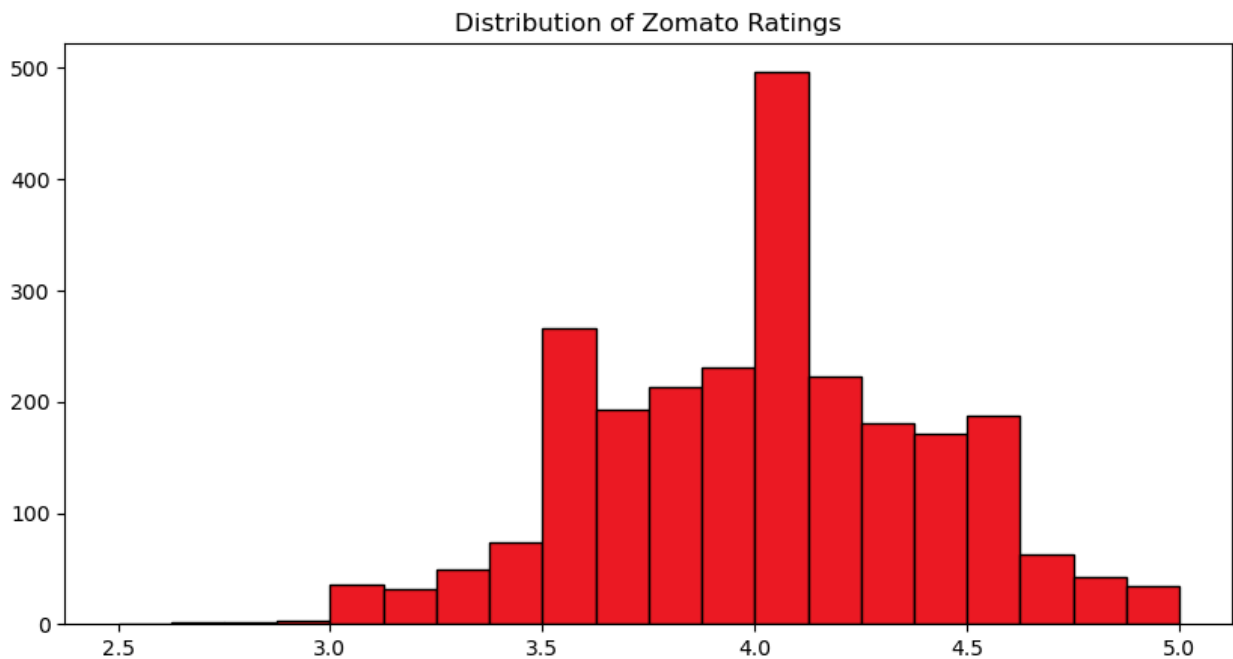
- Most restaurant ratings are concentrated between 3.5 and 4.5.
- Very few restaurants have extremely low ratings below 3.0.
- The distribution peaks around 4.0, indicating generally positive customer satisfaction across the platform.

```
In [15]: plt.figure(figsize=(10,5))

plt.hist(
    df['zomato_rating'],
    bins=20,
    color='#ED1C24',
    edgecolor='black'
)

plt.title("Distribution of Zomato Ratings")

plt.show()
```



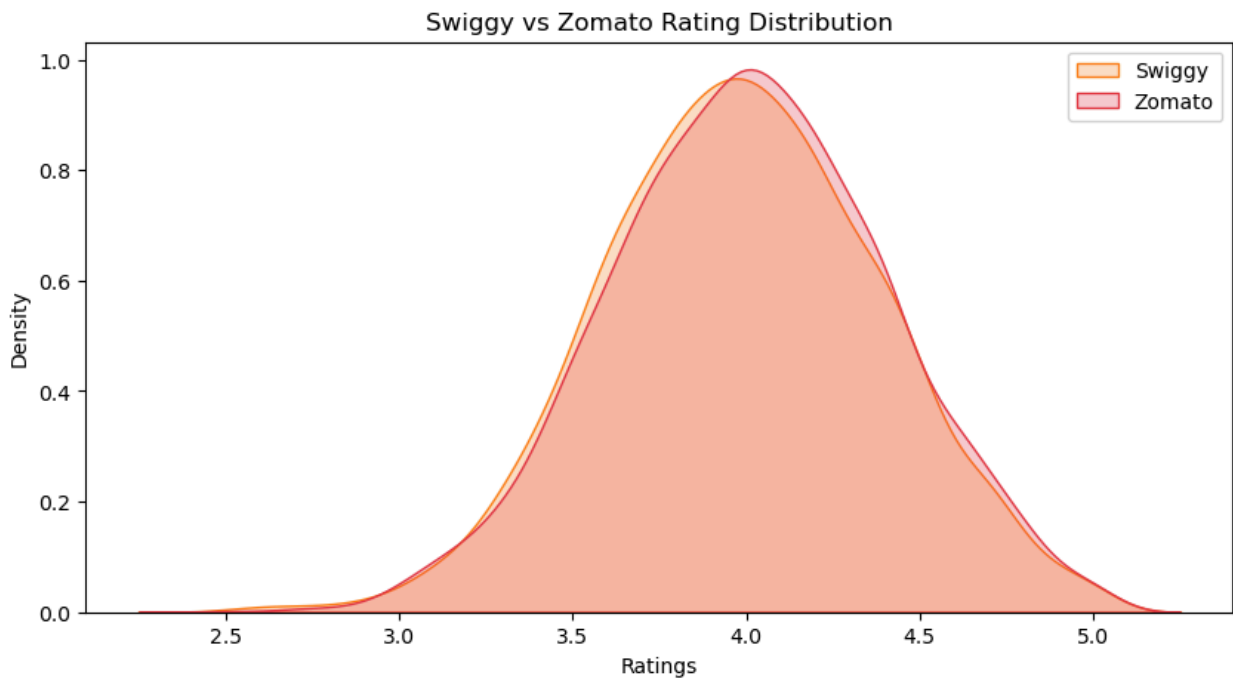
✕ Swiggy vs Zomato Rating Comparison

```
In [16]: plt.figure(figsize=(10,5))

sns.kdeplot(df['swiggy_rating'], label='Swiggy', color='#FC8019', fill=True)
sns.kdeplot(df['zomato_rating'], label='Zomato', color='#E23744', fill=True)

plt.title("Swiggy vs Zomato Rating Distribution")
plt.xlabel("Ratings")
plt.legend()

plt.show()
```



Observation

- Both Swiggy and Zomato exhibit highly similar rating distributions, indicating comparable customer satisfaction patterns across platforms.
- The majority of restaurant ratings are concentrated around the 4.0 range, suggesting generally positive user experiences.
- Extremely low ratings are relatively uncommon, which may reflect consistent service quality or typical rating inflation behavior observed in food delivery platforms.



Impact of Delivery Time on Customer Ratings

```
In [17]: # Swiggy Correlation
swiggy_corr = df['swiggy_avg_delivery_time_minutes'].corr(df['swiggy_rating'])

# Zomato Correlation
zomato_corr = df['zomato_avg_delivery_time_minutes'].corr(df['zomato_rating'])

print("Swiggy Correlation:", round(swiggy_corr, 3))
print("Zomato Correlation:", round(zomato_corr, 3))
```

Swiggy Correlation: -0.014
Zomato Correlation: 0.023

Observation

The correlation between delivery time and customer ratings is extremely weak for

both Swiggy (-0.014) and Zomato (0.023). This suggests that delivery speed alone does not significantly influence customer ratings within this dataset.

The findings indicate that customers may prioritize other factors such as food quality, pricing, packaging, and overall dining experience more heavily when rating restaurants on food delivery platforms.

💰 Impact of Discounts on Restaurant Profitability

```
In [18]: discount_profit = df.groupby('swiggy_discount_frequency_pct')['swiggy_estimate']

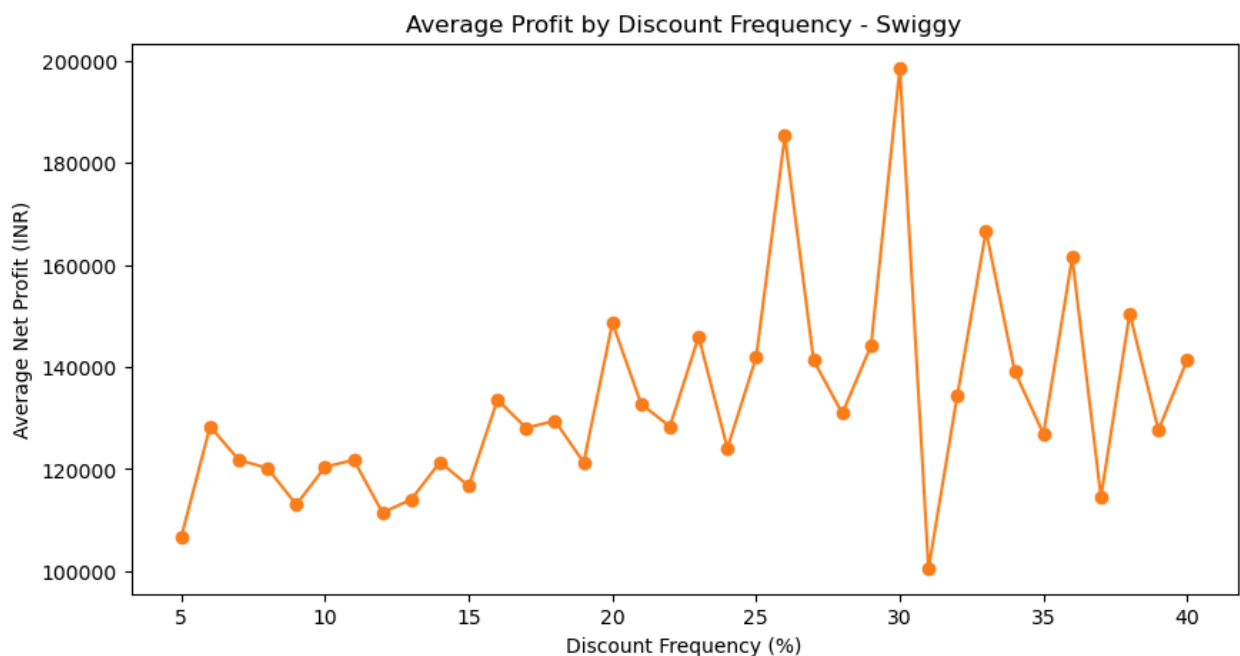
plt.figure(figsize=(10,5))

plt.plot(
    discount_profit.index,
    discount_profit.values,
    color='#FC8019',
    marker='o'
)

plt.xlabel("Discount Frequency (%)")
plt.ylabel("Average Net Profit (INR)")

plt.title("Average Profit by Discount Frequency - Swiggy")

plt.show()
```



```
In [19]: discount_profit_zomato = df.groupby('zomato_discount_frequency_pct')['zomato_e
```

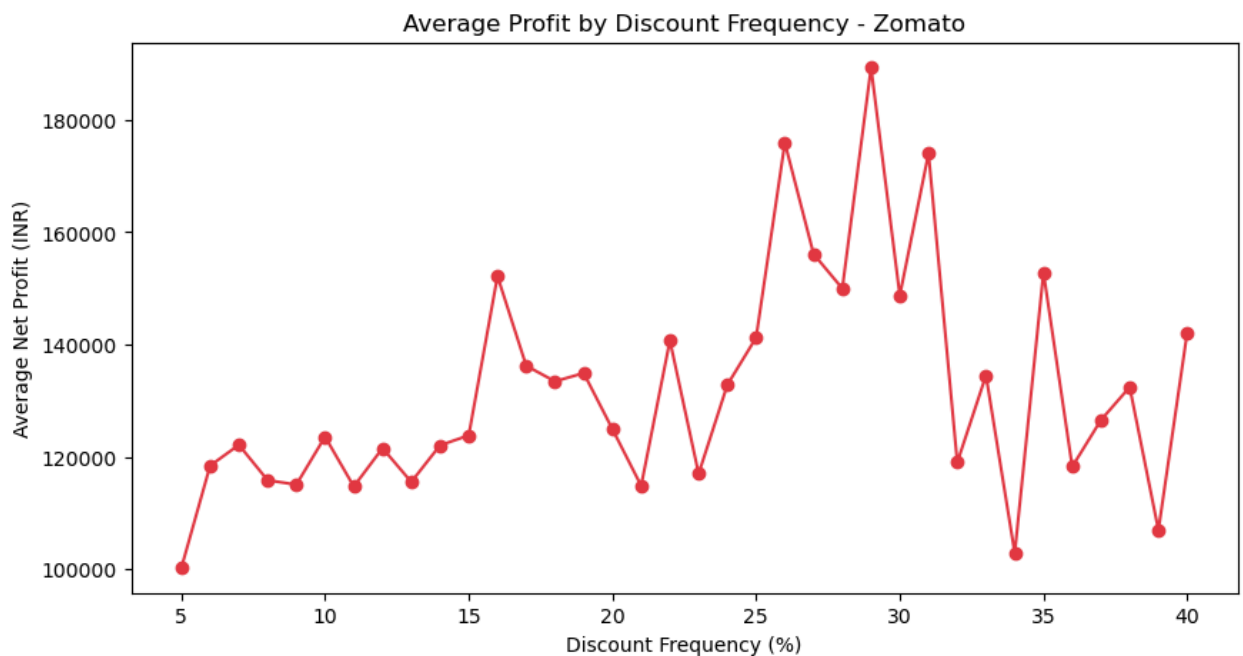
```
plt.figure(figsize=(10,5))

plt.plot(
    discount_profit_zomato.index,
    discount_profit_zomato.values,
    color='#E23744',
    marker='o'
)

plt.xlabel("Discount Frequency (%)")
plt.ylabel("Average Net Profit (INR)")

plt.title("Average Profit by Discount Frequency - Zomato")

plt.show()
```



Observation

- Both Swiggy and Zomato exhibit fluctuating profit trends across different discount frequency levels, indicating that discounts alone do not consistently determine restaurant profitability.
- Moderate to higher discount frequencies occasionally correspond with increased average profits, suggesting that strategic discounting may help boost customer acquisition and order volumes.
- However, the presence of sharp rises and drops in profitability across similar discount levels highlights that additional factors such as operational efficiency, pricing strategy, cuisine type, and customer demand likely play a significant role in determining overall profit performance.

- Overall, the analysis suggests that discounts can support growth when applied strategically, but excessive or poorly optimized discounting may not always guarantee higher profitability.



Relationship Between Orders and Revenue

```
In [20]: orders_revenue = df.groupby('swiggy_estimated_monthly_orders')['swiggy_estimated_monthly_revenue']

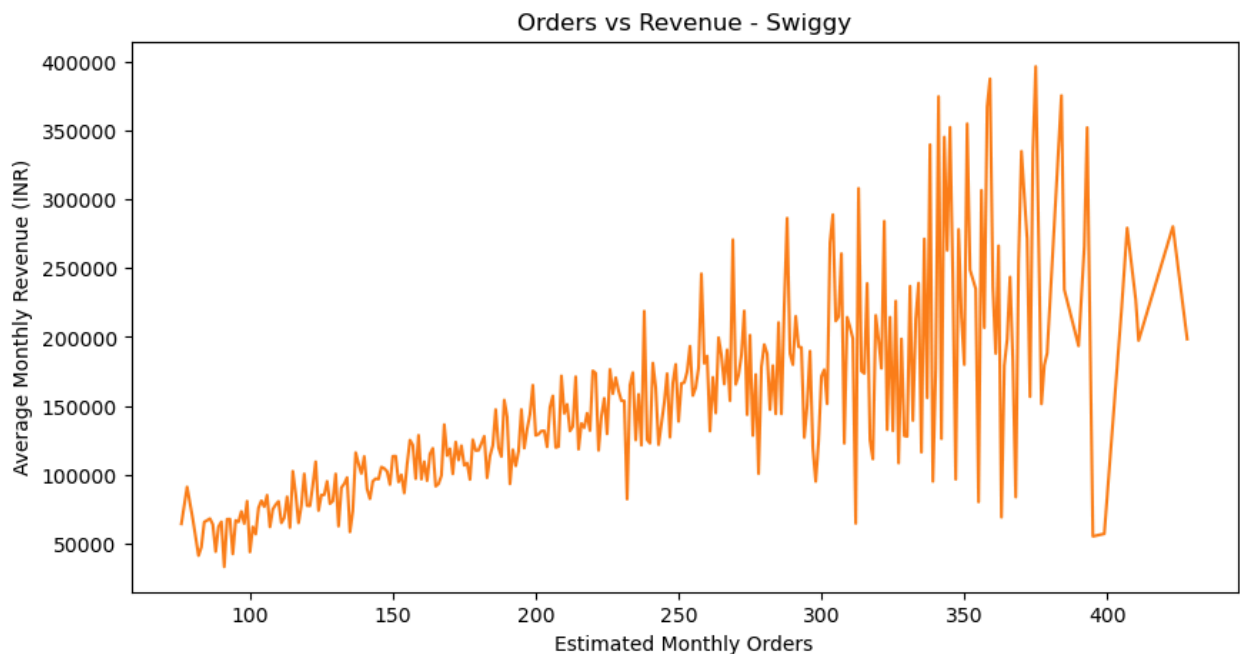
plt.figure(figsize=(10,5))

plt.plot(
    orders_revenue.index,
    orders_revenue.values,
    color='#FC8019'
)

plt.xlabel("Estimated Monthly Orders")
plt.ylabel("Average Monthly Revenue (INR)")

plt.title("Orders vs Revenue - Swiggy")

plt.show()
```



```
In [21]: orders_revenue_zomato = df.groupby('zomato_estimated_monthly_orders')['zomato_estimated_monthly_revenue']

plt.figure(figsize=(10,5))

plt.plot(
    orders_revenue_zomato.index,
    orders_revenue_zomato.values,
```

```

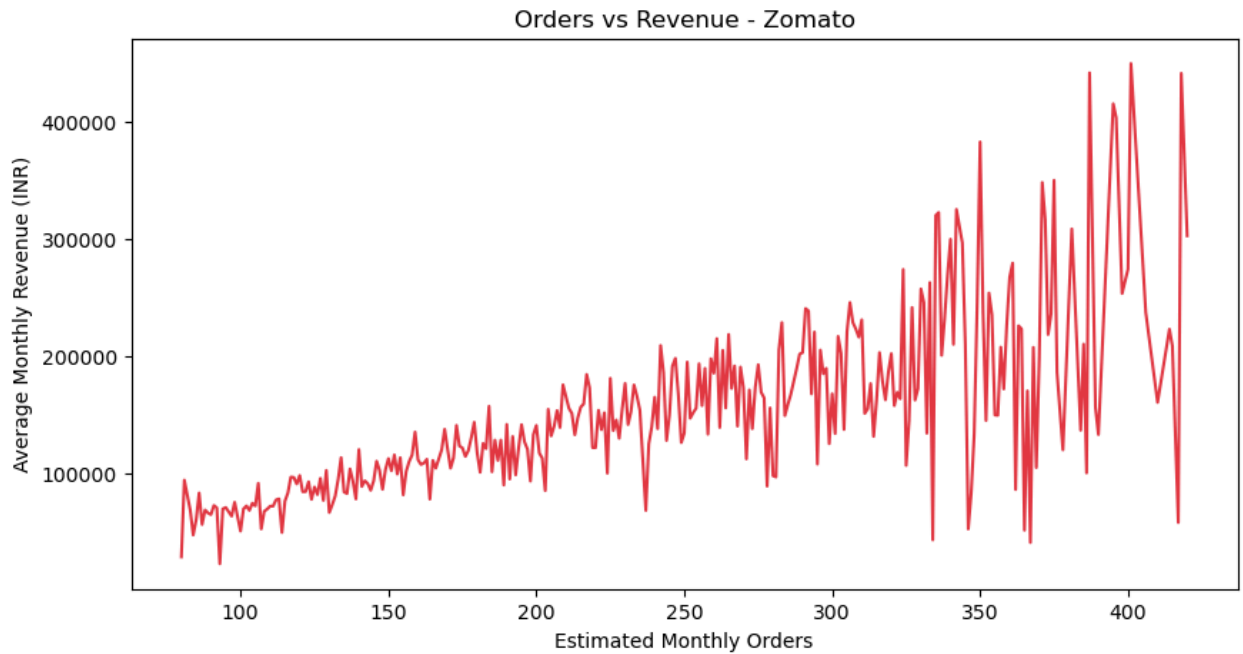
        color='#E23744'
    )

    plt.xlabel("Estimated Monthly Orders")
    plt.ylabel("Average Monthly Revenue (INR)")

    plt.title("Orders vs Revenue - Zomato")

    plt.show()

```



```

In [22]: import warnings
warnings.filterwarnings('ignore')

bins = [75, 150, 225, 300, 375, 450]

labels = [
    '75-150',
    '151-225',
    '226-300',
    '301-375',
    '376-450'
]

df['swiggy_order_bins'] = pd.cut(
    df['swiggy_estimated_monthly_orders'],
    bins=bins,
    labels=labels
)

# Average Revenue by Order Bin
orders_revenue = df.groupby('swiggy_order_bins')['swiggy_estimated_monthly_rev

# Plot

```

```
plt.figure(figsize=(10,5))

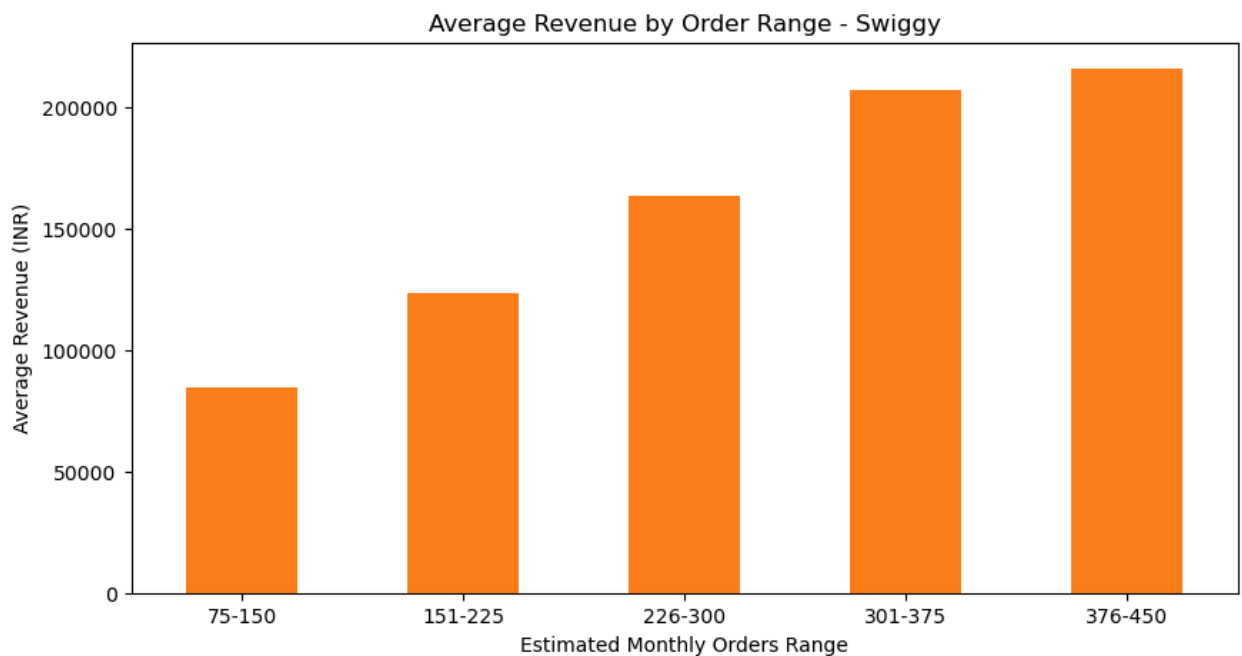
orders_revenue.plot(
    kind='bar',
    color='#FC8019'
)

plt.xlabel("Estimated Monthly Orders Range")
plt.ylabel("Average Revenue (INR)")

plt.title("Average Revenue by Order Range - Swiggy")

plt.xticks(rotation=0)

plt.show()
```



```
In [23]: # Creating Readable Order Bins

bins = [75, 150, 225, 300, 375, 450]

labels = [
    '75-150',
    '151-225',
    '226-300',
    '301-375',
    '376-450'
]

df['zomato_order_bins'] = pd.cut(
    df['zomato_estimated_monthly_orders'],
    bins=bins,
    labels=labels
)
```



```

# Average Revenue by Order Bin
orders_revenue_zomato = df.groupby('zomato_order_bins')['zomato_estimated_monthly_revenue'].mean()

# Plot
plt.figure(figsize=(10,5))

orders_revenue_zomato.plot(
    kind='bar',
    color='#E23744'
)

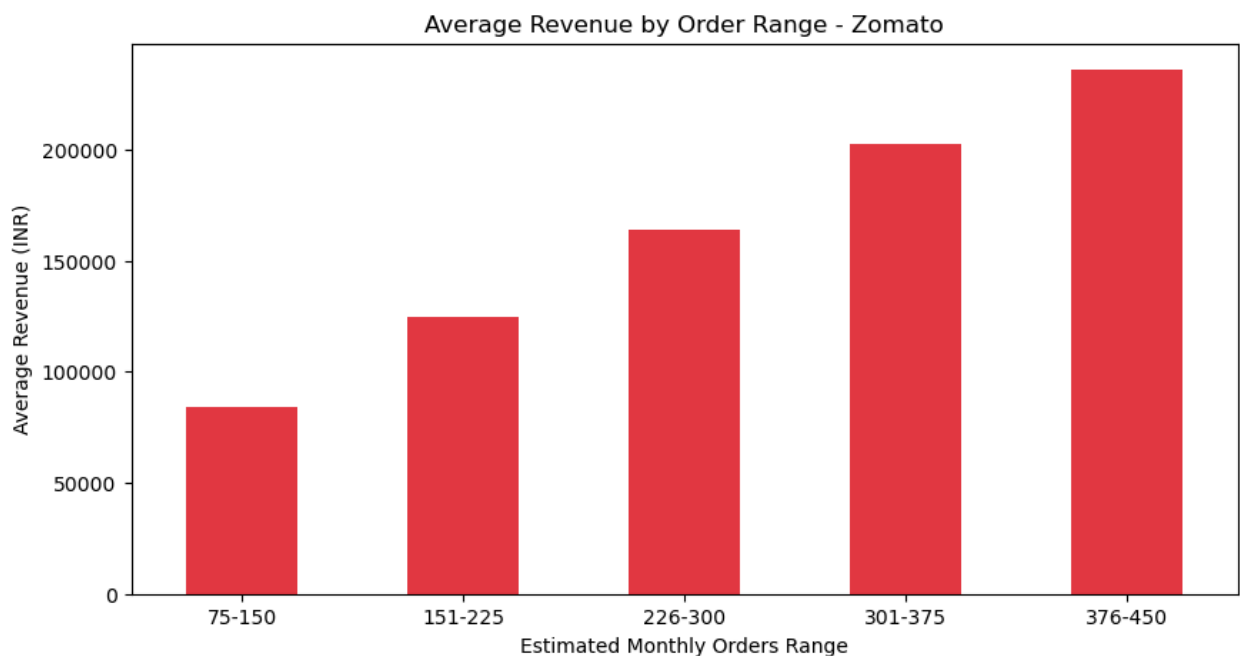
plt.xlabel("Estimated Monthly Orders Range")
plt.ylabel("Average Revenue (INR)")

plt.title("Average Revenue by Order Range - Zomato")

plt.xticks(rotation=0)

plt.show()

```



Observation

- Both Swiggy and Zomato demonstrate a clear positive relationship between order volume and average monthly revenue.
- Restaurants receiving higher monthly orders consistently generate significantly higher revenue across both platforms.
- The revenue growth pattern appears steady and scalable, indicating that increasing customer order volume is a major driver of business growth in food delivery platforms.

- Zomato shows slightly higher average revenue in the highest order range category, suggesting stronger revenue generation at larger order volumes within this dataset.
- Overall, the analysis highlights that order volume plays a critical role in determining restaurant revenue performance across both platforms.



City-wise Revenue Analysis

```
In [24]: swiggy_city_revenue = df.groupby('city')['swiggy_estimated_monthly_revenue_inr']

plt.figure(figsize=(12,6))

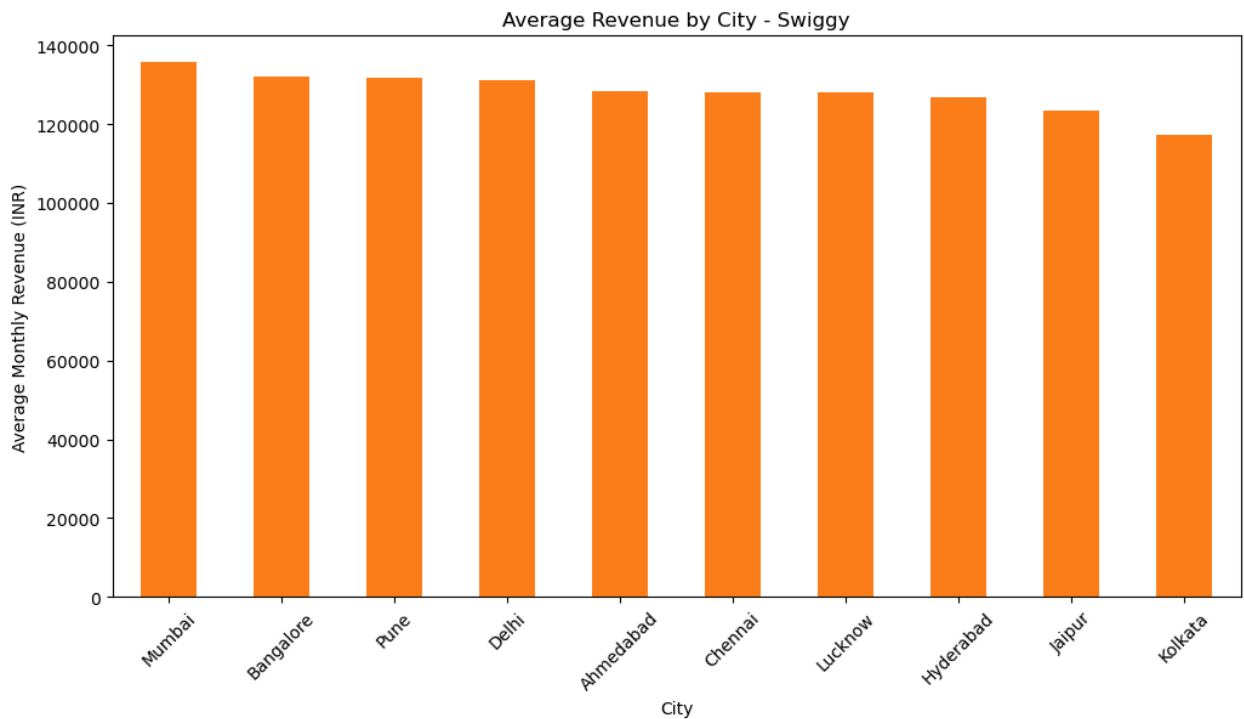
swiggy_city_revenue.plot(
    kind='bar',
    color='#FC8019'
)

plt.xlabel("City")
plt.ylabel("Average Monthly Revenue (INR)")

plt.title("Average Revenue by City - Swiggy")

plt.xticks(rotation=45)

plt.show()
```



```
In [25]: zomato_city_revenue = df.groupby('city')['zomato_estimated_monthly_revenue_inr']
```

```
plt.figure(figsize=(12,6))

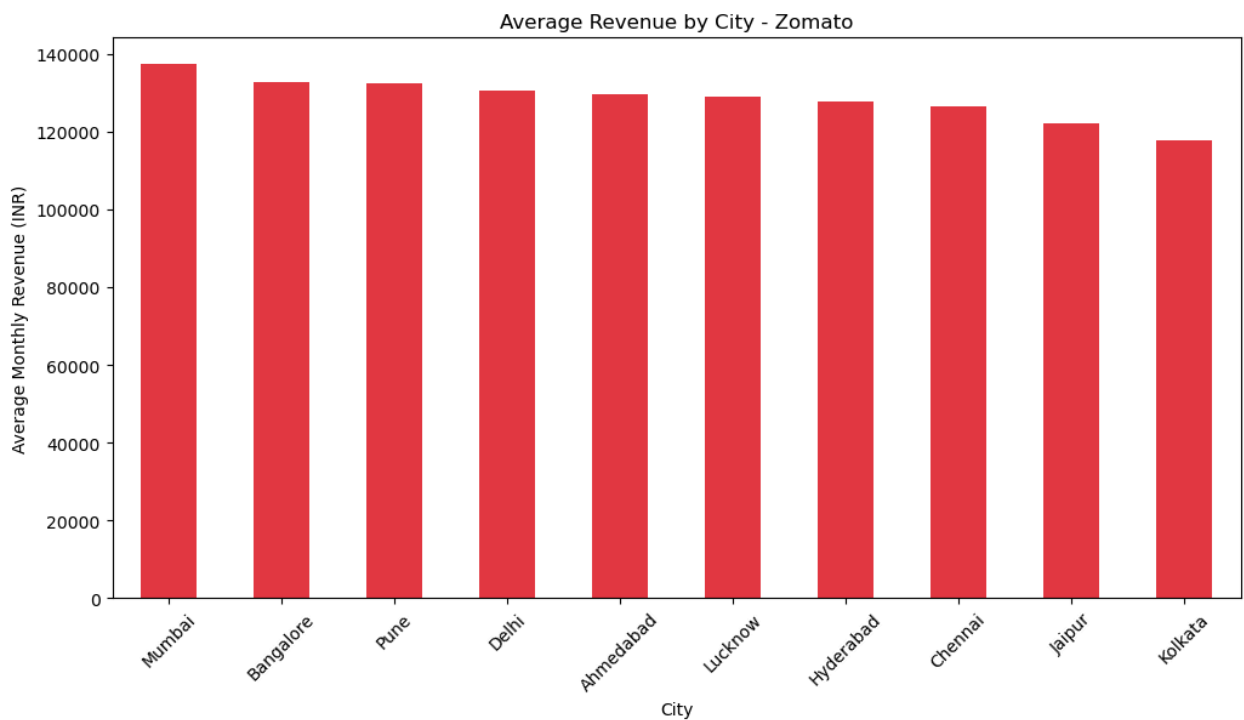
zomato_city_revenue.plot(
    kind='bar',
    color='#E23744'
)

plt.xlabel("City")
plt.ylabel("Average Monthly Revenue (INR)")

plt.title("Average Revenue by City - Zomato")

plt.xticks(rotation=45)

plt.show()
```



Observation

- Revenue distribution across cities appears relatively balanced for both Swiggy and Zomato, with no extreme dominance observed from any single city.
- Mumbai, Bangalore, and Pune demonstrate slightly higher average revenue generation compared to other cities across both platforms.
- The similarity in revenue patterns suggests that both platforms maintain comparable market penetration and customer demand across major Indian cities.
- Overall, the analysis indicates a fairly diversified geographic revenue

distribution within the dataset.



Most Profitable Cuisines

```
In [26]: swiggy_cuisine_profit = df.groupby('cuisines')['swiggy_estimated_net_profit_inr']

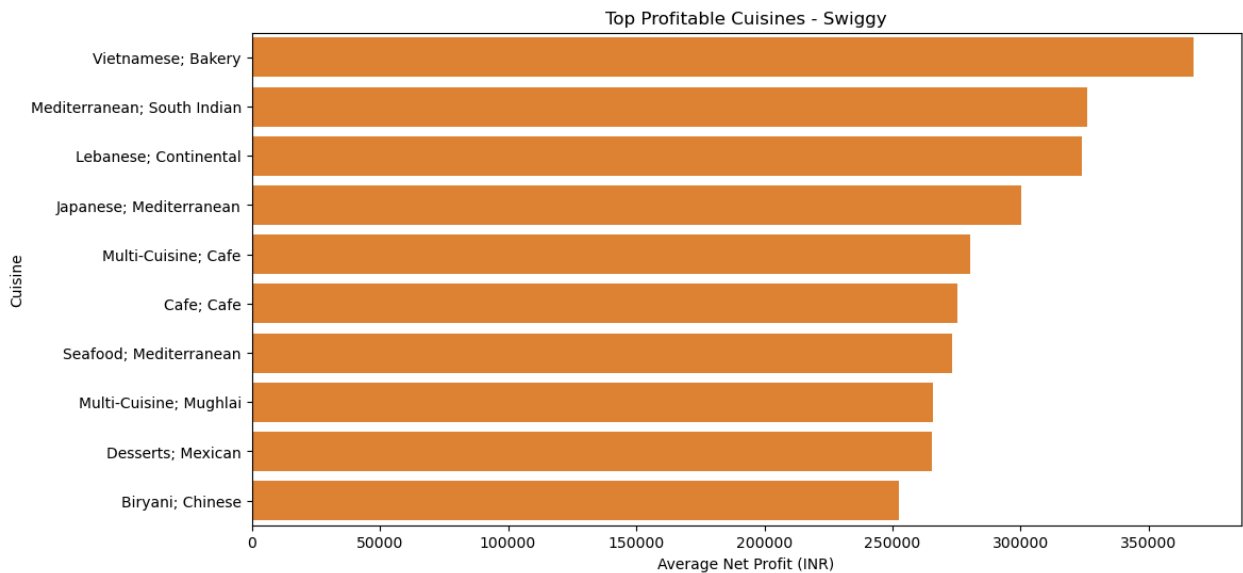
plt.figure(figsize=(12,6))

sns.barplot(
    x=swiggy_cuisine_profit.values,
    y=swiggy_cuisine_profit.index,
    color='#FC8019'
)

plt.xlabel("Average Net Profit (INR)")
plt.ylabel("Cuisine")

plt.title("Top Profitable Cuisines - Swiggy")

plt.show()
```



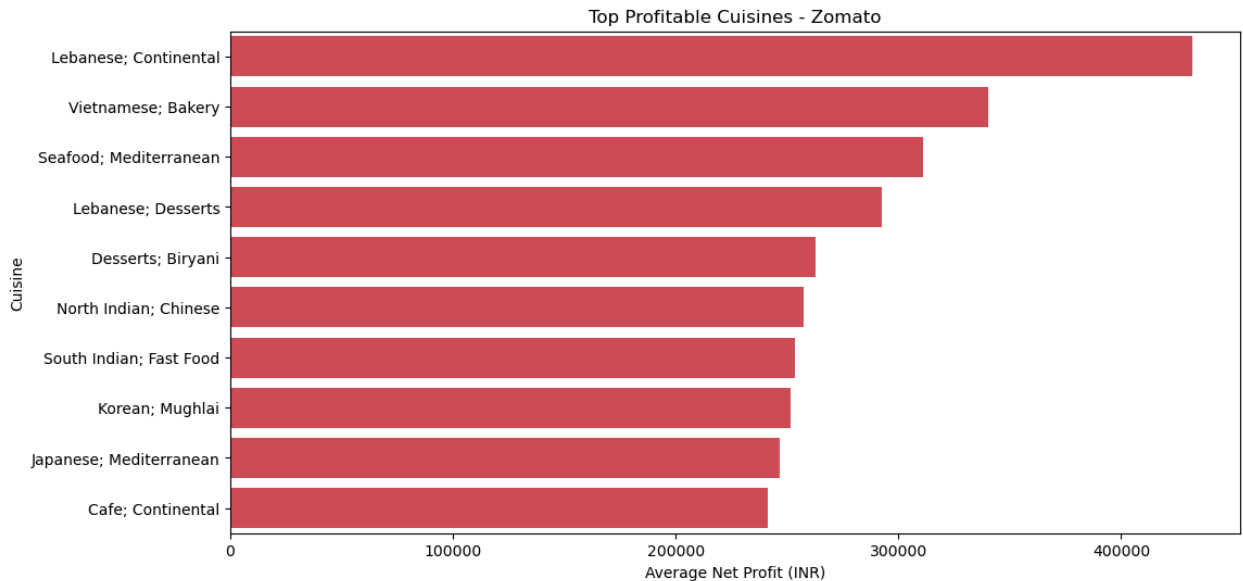
```
In [27]: zomato_cuisine_profit = df.groupby('cuisines')['zomato_estimated_net_profit_inr']

plt.figure(figsize=(12,6))

sns.barplot(
    x=zomato_cuisine_profit.values,
    y=zomato_cuisine_profit.index,
    color='#E23744'
)

plt.xlabel("Average Net Profit (INR)")
plt.ylabel("Cuisine")
```

```
plt.title("Top Profitable Cuisines - Zomato")
plt.show()
```



Observation

- Cuisine profitability patterns vary noticeably between Swiggy and Zomato, indicating differences in customer preferences and platform performance across cuisine categories.
- Swiggy demonstrates strong profitability in cuisine combinations such as Vietnamese & Bakery, Mediterranean & South Indian, and Japanese & Mediterranean.
- Zomato shows comparatively higher profitability in Lebanese & Continental, Seafood & Mediterranean, and Lebanese & Desserts cuisine combinations.
- Certain cuisine categories such as Japanese & Mediterranean and Seafood & Mediterranean appear consistently profitable across both platforms, suggesting strong and stable customer demand.
- The variation in cuisine profitability highlights potential opportunities for platform-specific cuisine expansion, targeted restaurant partnerships, and strategic marketing initiatives.

✕ Platform Performance Comparison

```
In [28]: platform_counts = df['platform_performance_better'].value_counts()
plt.figure(figsize=(8,5))
```

```

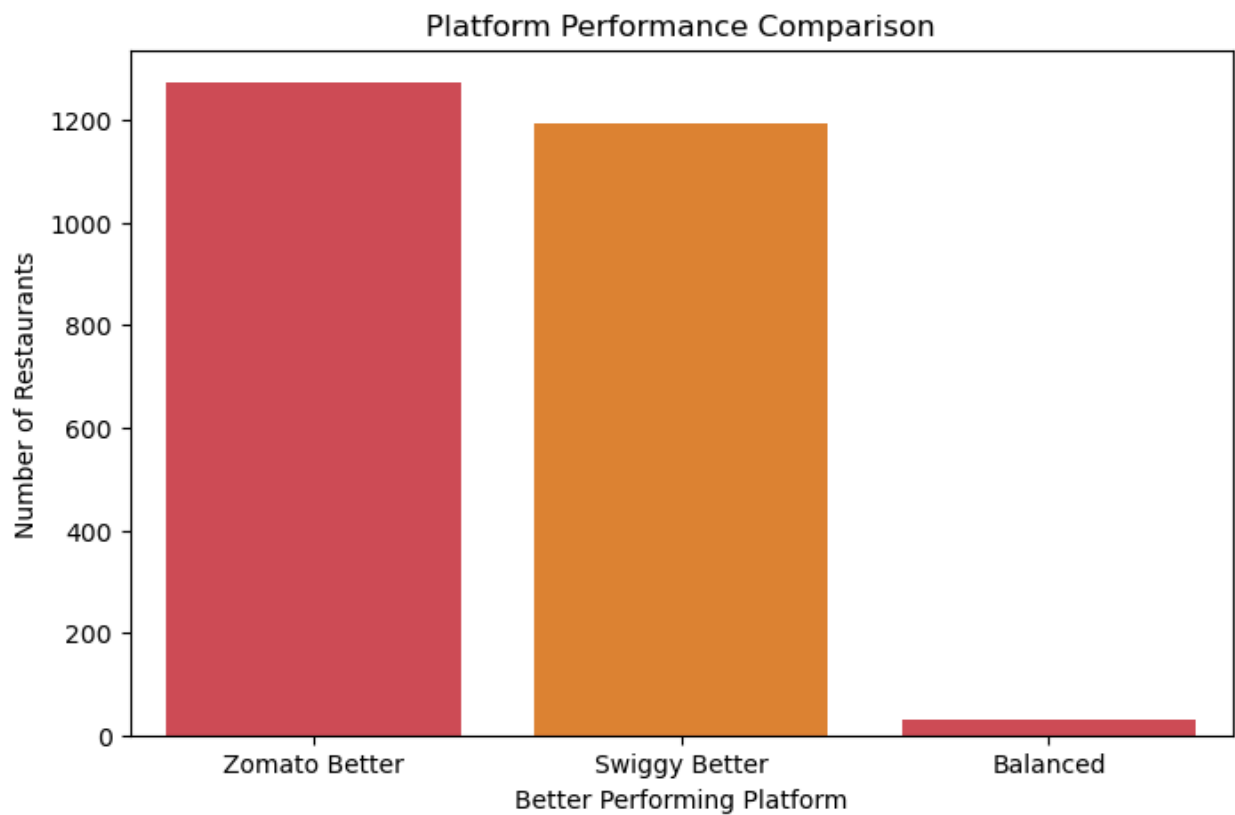
sns.barplot(
    x=platform_counts.index,
    y=platform_counts.values,
    palette=['#E23744', '#FC8019']
)

plt.xlabel("Better Performing Platform")
plt.ylabel("Number of Restaurants")

plt.title("Platform Performance Comparison")

plt.show()

```



Observation

- Zomato appears to outperform Swiggy across a slightly higher number of restaurants within the dataset.
- Swiggy remains highly competitive, with overall platform performance distribution remaining relatively balanced between both platforms.
- Very few restaurants demonstrate completely balanced platform performance, suggesting that restaurants generally tend to perform better on one platform over the other.
- The competitive closeness between Swiggy and Zomato highlights the intense rivalry within the Indian food delivery ecosystem.

★ Impact of Customer Ratings on Revenue

```
In [29]: # Creating Rating Categories

bins = [2.5, 3.5, 4.0, 4.5, 5.0]

labels = [
    'Low Rated',
    'Average Rated',
    'High Rated',
    'Top Rated'
]

df['swiggy_rating_group'] = pd.cut(
    df['swiggy_rating'],
    bins=bins,
    labels=labels
)

# Average Revenue by Rating Group
swiggy_rating_revenue = df.groupby('swiggy_rating_group')['swiggy_estimated_monetary_value'].mean()

# Plot
plt.figure(figsize=(10,5))

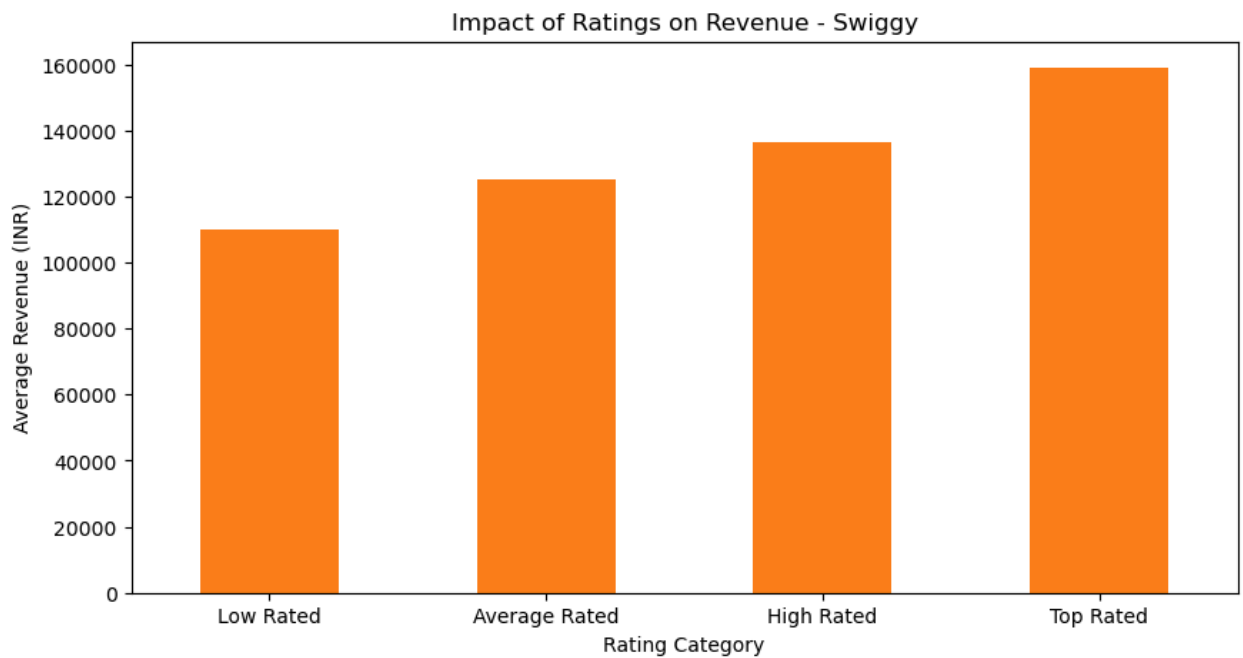
swiggy_rating_revenue.plot(
    kind='bar',
    color='#FC8019'
)

plt.xlabel("Rating Category")
plt.ylabel("Average Revenue (INR)")

plt.title("Impact of Ratings on Revenue - Swiggy")

plt.xticks(rotation=0)

plt.show()
```



```
In [30]: # Creating Rating Categories

df['zomato_rating_group'] = pd.cut(
    df['zomato_rating'],
    bins=bins,
    labels=labels
)

# Average Revenue by Rating Group
zomato_rating_revenue = df.groupby('zomato_rating_group')['zomato_estimated_monthly_revenue'].mean()

# Plot
plt.figure(figsize=(10,5))

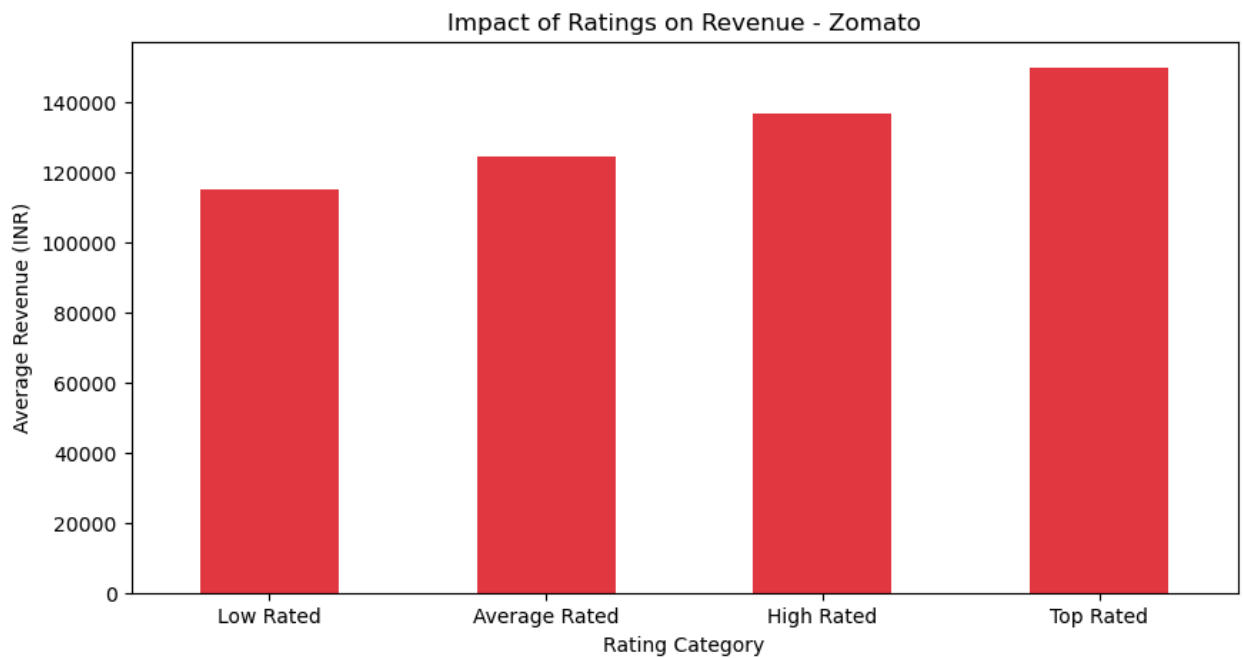
zomato_rating_revenue.plot(
    kind='bar',
    color='#E23744'
)

plt.xlabel("Rating Category")
plt.ylabel("Average Revenue (INR)")

plt.title("Impact of Ratings on Revenue - Zomato")

plt.xticks(rotation=0)

plt.show()
```

💰 Impact of Customer Ratings on Profitability

```
In [31]: # Average Profit by Rating Group
swiggy_rating_profit = df.groupby('swiggy_rating_group')['swiggy_estimated_net

# Plot
plt.figure(figsize=(10,5))

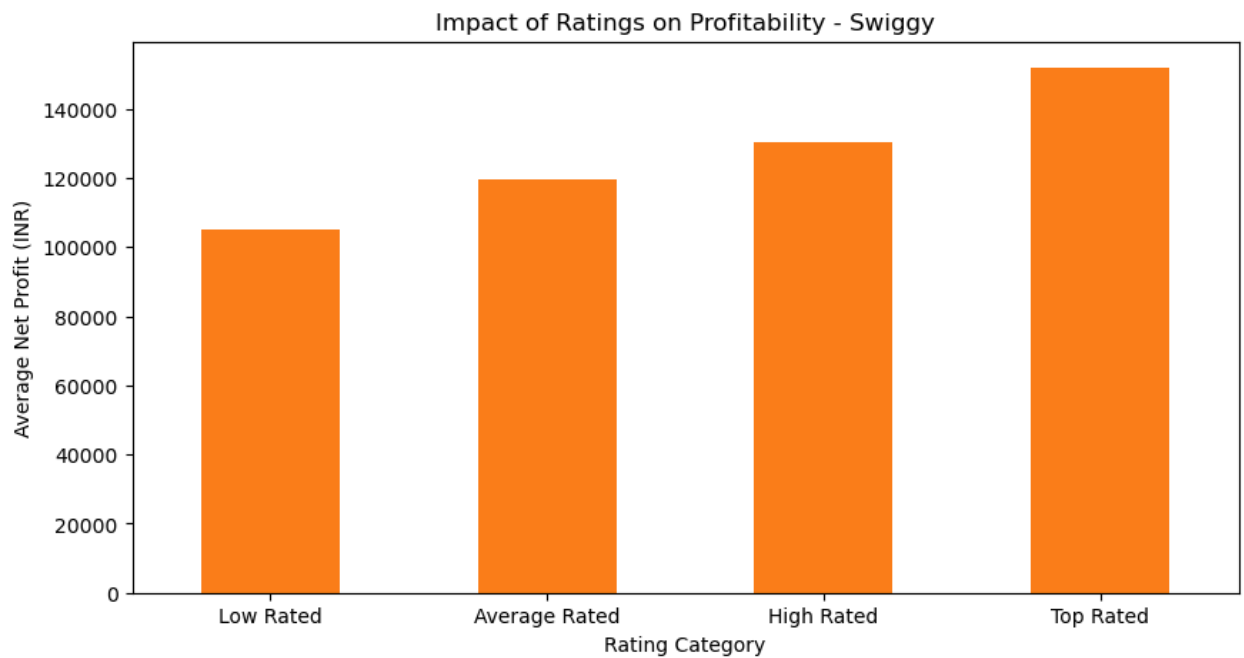
swiggy_rating_profit.plot(
    kind='bar',
    color='#FC8019'
)

plt.xlabel("Rating Category")
plt.ylabel("Average Net Profit (INR)")

plt.title("Impact of Ratings on Profitability - Swiggy")

plt.xticks(rotation=0)

plt.show()
```



```
In [32]: # Average Profit by Rating Group
zomato_rating_profit = df.groupby('zomato_rating_group')['zomato_estimated_net

# Plot
plt.figure(figsize=(10,5))

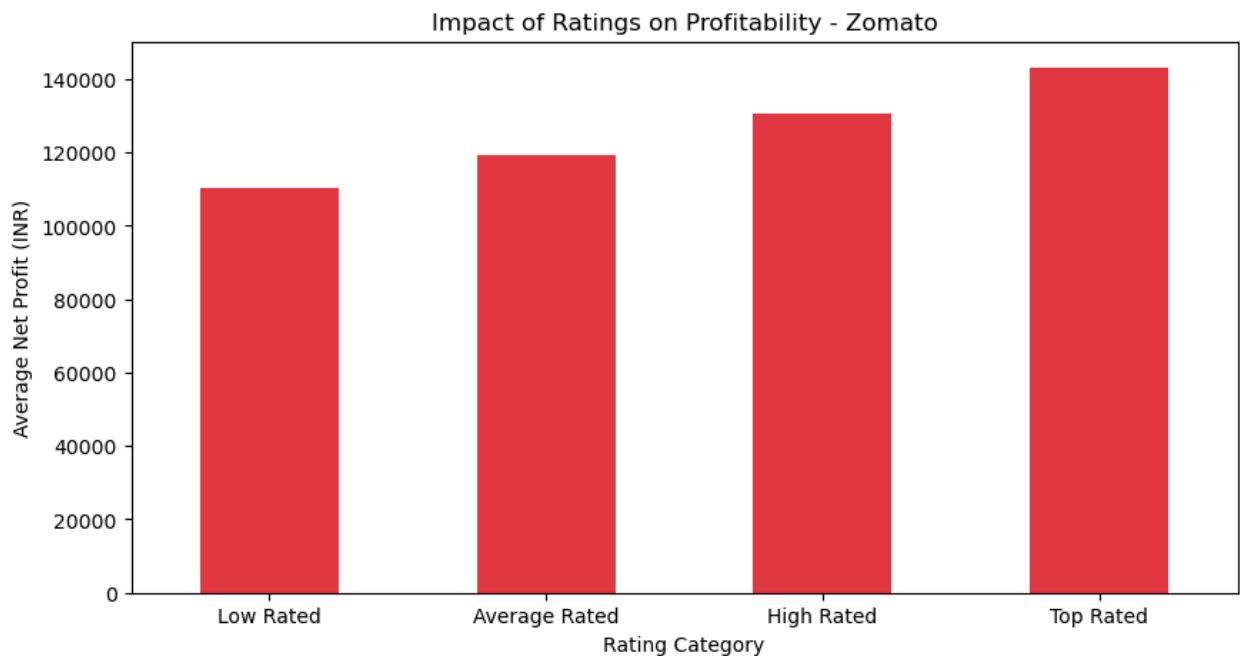
zomato_rating_profit.plot(
    kind='bar',
    color='#E23744'
)

plt.xlabel("Rating Category")
plt.ylabel("Average Net Profit (INR)")

plt.title("Impact of Ratings on Profitability - Zomato")

plt.xticks(rotation=0)

plt.show()
```



Observation

- Restaurants with higher customer ratings consistently demonstrate stronger profitability across both Swiggy and Zomato.
- Top-rated restaurants generate significantly higher average net profits compared to low-rated restaurants, indicating a strong relationship between customer satisfaction and financial performance.
- The findings suggest that maintaining food quality, service consistency, and positive customer experience can directly contribute to higher business profitability.
- Customer ratings appear to influence not only platform visibility and customer trust but also long-term revenue and profit generation.

Correlation Analysis of Business Metrics

In [33]: *# Selecting Numerical Columns*

```
corr_columns = [  
    'swiggy_rating',  
    'zomato_rating',  
    'swiggy_avg_delivery_time_minutes',  
    'zomato_avg_delivery_time_minutes',  
    'swiggy_estimated_monthly_orders',  
    'zomato_estimated_monthly_orders',  
    'swiggy_estimated_monthly_revenue_inr',  
    'zomato_estimated_monthly_revenue_inr',  
    'swiggy_estimated_net_profit_inr',  
]
```

```

        'zomato_estimated_net_profit_inr',
        'swiggy_discount_frequency_pct',
        'zomato_discount_frequency_pct'
    ]

    # Correlation Matrix
    corr_matrix = df[corr_columns].corr()

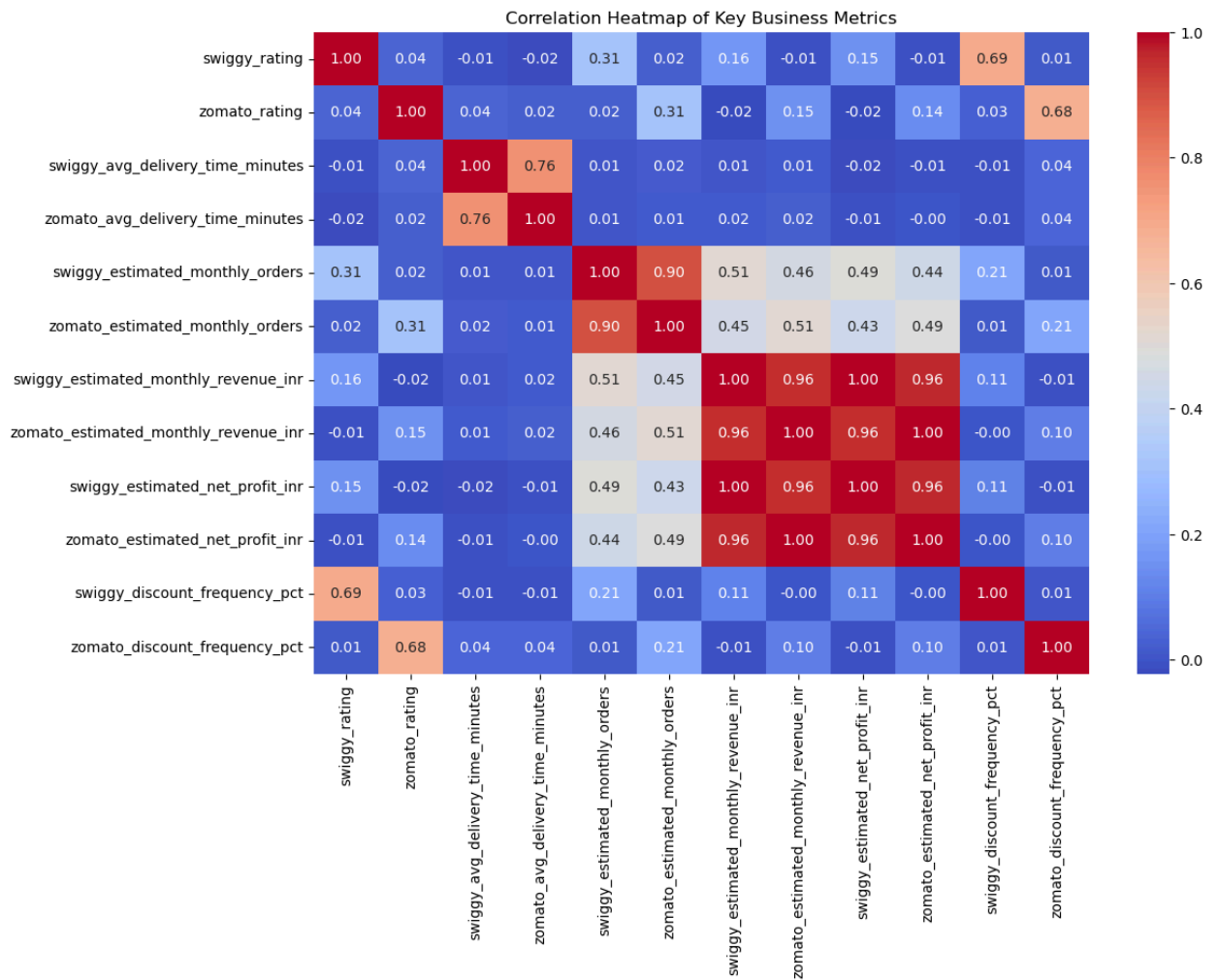
    # Plot
    plt.figure(figsize=(12,8))

    sns.heatmap(
        corr_matrix,
        annot=True,
        cmap='coolwarm',
        fmt='.2f'
    )

    plt.title("Correlation Heatmap of Key Business Metrics")

    plt.show()

```



Observation

- Revenue and profitability exhibit extremely strong positive correlations across both Swiggy and Zomato, indicating that restaurants generating higher revenue also tend to achieve stronger financial performance.
- Monthly order volume demonstrates a moderate positive relationship with both revenue and profit, suggesting that higher customer demand contributes significantly to restaurant growth.
- Delivery time shows almost no meaningful correlation with customer ratings, reinforcing earlier findings that customer satisfaction may depend more heavily on food quality, pricing, and overall service experience.
- Discount frequency displays a notable positive relationship with ratings, indicating that promotional strategies and customer value perception may influence customer satisfaction levels.
- Strong correlations between Swiggy and Zomato operational metrics suggest that restaurant performance patterns remain relatively consistent across both platforms.



High-Rated but Low-Profit Restaurants

```
In [34]: # Filtering Restaurants

high_rating_low_profit = df[
    (df['swiggy_rating'] >= 4.5) &
    (df['swiggy_estimated_net_profit_inr'] < df['swiggy_estimated_net_profit_i
]

# Top Results
high_rating_low_profit[
    [
        'restaurant_name',
        'cuisines',
        'city',
        'swiggy_rating',
        'swiggy_estimated_net_profit_inr'
    ]
].head(10)
```

Out[34]:	restaurant_name	cuisines	city	swiggy_rating	swiggy_estimated_net
39	Restaurant_40	North Indian	Mumbai	4.6	
55	Restaurant_56	South Indian	Mumbai	4.5	
59	Restaurant_60	Multi-Cuisine	Mumbai	4.7	
61	Restaurant_62	Continental	Mumbai	4.5	
79	Restaurant_80	Lebanese	Mumbai	4.9	
104	Restaurant_105	Thai	Mumbai	4.7	
153	Restaurant_154	Continental	Mumbai	4.5	
179	Restaurant_180	Fast Food	Mumbai	4.7	
235	Restaurant_236	Street Food; Vietnamese	Mumbai	4.7	
247	Restaurant_248	Multi-Cuisine; South Indian	Mumbai	4.5	

```
In [35]: # Filtering Restaurants

high_rating_low_profit_zomato = df[
    (df['zomato_rating'] >= 4.5) &
    (df['zomato_estimated_net_profit_inr'] < df['zomato_estimated_net_profit_i
]

# Top Results
high_rating_low_profit_zomato[
    [
        'restaurant_name',
        'cuisines',
        'city',
        'zomato_rating',
        'zomato_estimated_net_profit_inr'
    ]
].head(10)
```

Out[35]:

	restaurant_name	cuisines	city	zomato_rating	zomato_estimated
8	Restaurant_9	Mediterranean; Fast Food	Mumbai	4.6	
39	Restaurant_40	North Indian	Mumbai	4.5	
49	Restaurant_50	Japanese; Seafood	Mumbai	4.5	
79	Restaurant_80	Lebanese	Mumbai	4.7	
103	Restaurant_104	Cafe	Mumbai	4.6	
118	Restaurant_119	Pan-Asian; Pan-Asian	Mumbai	4.8	
147	Restaurant_148	Vietnamese	Mumbai	4.8	
149	Restaurant_150	Japanese	Mumbai	4.8	
171	Restaurant_172	Italian; Cafe	Mumbai	4.6	
189	Restaurant_190	Korean; Japanese	Mumbai	4.5	

```
In [39]: # Correlation with Swiggy Profit
# Selecting Only Swiggy Metrics

swiggy_cols = [
    'swiggy_rating',
    'swiggy_total_reviews',
    'swiggy_delivery_fee_inr',
    'swiggy_avg_delivery_time_minutes',
    'swiggy_platform_commission_pct',
    'swiggy_discount_frequency_pct',
    'swiggy_estimated_monthly_orders',
    'swiggy_estimated_monthly_revenue_inr',
    'swiggy_estimated_net_profit_inr'
]

# Correlation with Profit
swiggy_profit_corr = df[swiggy_cols].corr()['swiggy_estimated_net_profit_inr']

# Removing self-correlation
swiggy_profit_corr = swiggy_profit_corr.drop('swiggy_estimated_net_profit_inr')

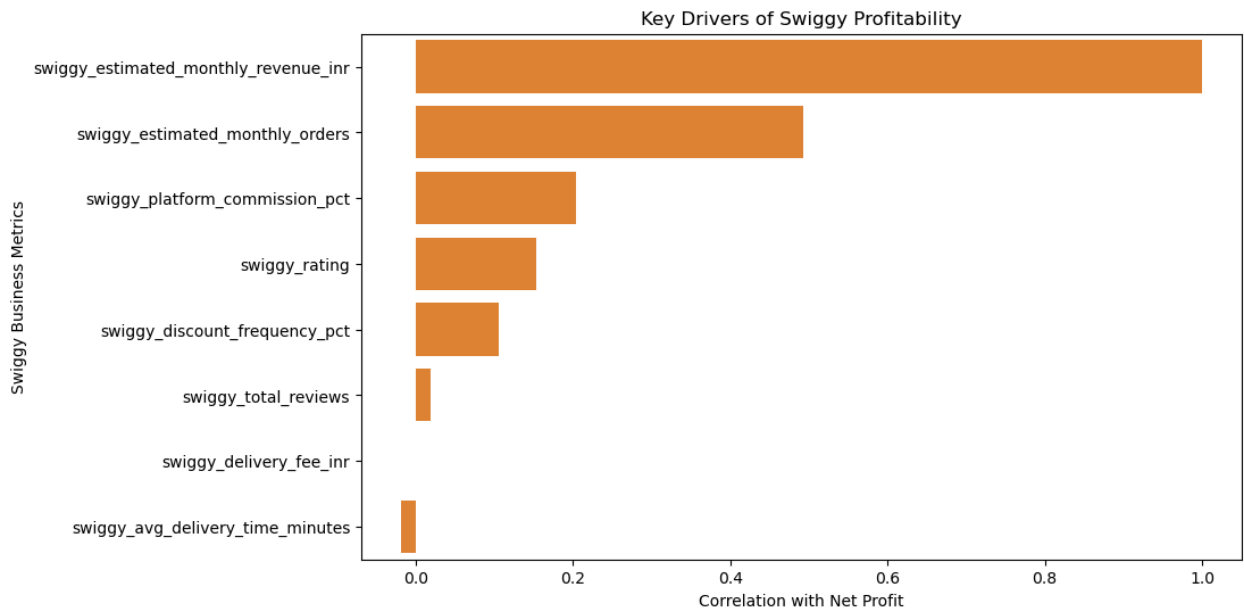
# Plot
plt.figure(figsize=(10,6))

sns.barplot(
    x=swiggy_profit_corr.values,
    y=swiggy_profit_corr.index,
    color='#FC8019'
)
```

```
plt.xlabel("Correlation with Net Profit")
plt.ylabel("Swiggy Business Metrics")

plt.title("Key Drivers of Swiggy Profitability")

plt.show()
```



In [40]: *# Selecting Only Zomato Metrics*

```
zomato_cols = [
    'zomato_rating',
    'zomato_total_reviews',
    'zomato_delivery_fee_inr',
    'zomato_avg_delivery_time_minutes',
    'zomato_platform_commission_pct',
    'zomato_discount_frequency_pct',
    'zomato_estimated_monthly_orders',
    'zomato_estimated_monthly_revenue_inr',
    'zomato_estimated_net_profit_inr'
]

# Correlation with Profit
zomato_profit_corr = df[zomato_cols].corr()['zomato_estimated_net_profit_inr']

# Removing self-correlation
zomato_profit_corr = zomato_profit_corr.drop('zomato_estimated_net_profit_inr')

# Plot
plt.figure(figsize=(10,6))

sns.barplot(
    x=zomato_profit_corr.values,
    y=zomato_profit_corr.index,
    color='#E23744'
```



```

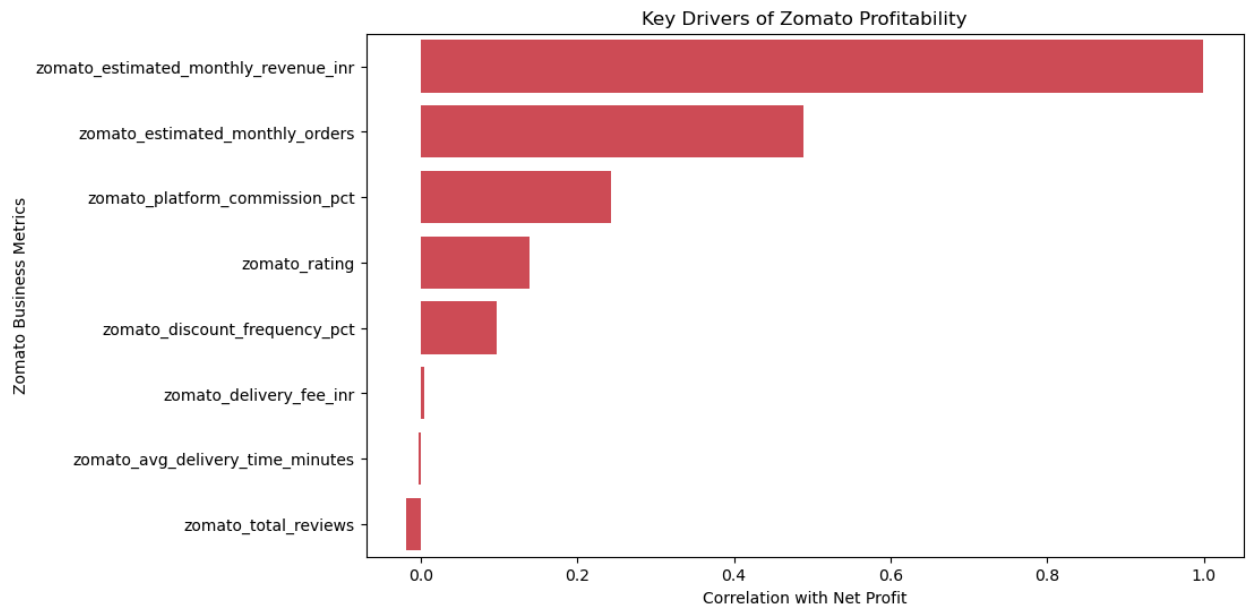
)

plt.xlabel("Correlation with Net Profit")
plt.ylabel("Zomato Business Metrics")

plt.title("Key Drivers of Zomato Profitability")

plt.show()

```



Conclusion

The profitability driver analysis highlights that revenue and monthly order volume are the strongest contributors to restaurant profitability across both Swiggy and Zomato. Restaurants generating higher customer demand and stronger revenue streams tend to achieve significantly better financial performance.

Customer ratings also demonstrate a positive relationship with profitability, indicating that customer satisfaction, food quality, and overall service experience contribute meaningfully to long-term business success.

Interestingly, discount frequency shows a moderate positive relationship with profitability, suggesting that strategic promotional campaigns may help improve customer acquisition and sales performance when managed effectively.

On the other hand, delivery time and total review count exhibit very weak or negligible relationships with profitability, indicating that operational speed and review quantity alone may not significantly influence financial outcomes within this dataset.

Overall, the analysis suggests that sustainable restaurant profitability is primarily

driven by strong revenue generation, consistent customer demand, and positive customer experience rather than operational metrics alone.

Overall Conclusion & Key Business Insights

This exploratory data analysis provided a comparative business and operational overview of restaurants operating across Swiggy and Zomato within the Indian food delivery ecosystem. The analysis focused on customer behavior, profitability patterns, platform performance, operational metrics, and strategic business drivers.

Key Insights

Customer Ratings Significantly Influence Profitability

Restaurants with higher customer ratings consistently demonstrated stronger revenue generation and higher profitability across both platforms. The analysis suggests that customer satisfaction, food quality, and service consistency play a critical role in long-term restaurant success.

Revenue & Order Volume Are the Strongest Drivers of Profit

Correlation analysis revealed that estimated monthly revenue and order volume exhibit the strongest positive relationship with restaurant profitability. Restaurants generating higher customer demand were consistently associated with stronger financial performance and operational scalability.

Cuisine Preferences Vary Across Platforms

Cuisine profitability analysis highlighted noticeable differences between Swiggy and Zomato. Certain cuisine combinations such as Vietnamese & Bakery and Mediterranean-based cuisines performed exceptionally well on Swiggy, while Lebanese & Continental combinations demonstrated stronger profitability on Zomato. This suggests that customer preferences and cuisine performance vary across platforms.

✂ Zomato Slightly Outperformed Swiggy Overall

Platform performance comparison indicated that Zomato marginally outperformed Swiggy across a larger number of restaurants within the dataset. However, both platforms demonstrated highly competitive and closely balanced market performance overall.

Strategic Discounts Can Improve Business Performance

Discount frequency showed a moderate positive relationship with ratings and profitability, indicating that carefully optimized promotional campaigns may improve customer engagement, increase ordering activity, and positively influence revenue generation.

Delivery Time Had Limited Impact on Profitability

Operational metrics such as average delivery time demonstrated minimal influence on ratings and profitability. This suggests that customers may prioritize food quality, customer experience, pricing, and overall value perception more heavily than delivery speed alone.

Restaurant Performance Remains Consistent Across Platforms

Strong correlations between Swiggy and Zomato operational metrics indicate that restaurant performance patterns remain relatively stable across both platforms. Restaurants performing well on one platform often demonstrated comparable performance on the other.

Final Business Understanding

The analysis overall suggests that sustainable restaurant profitability within food delivery platforms depends primarily on:

- Strong customer satisfaction

- Higher order volumes
- Revenue scalability
- Strategic promotional activities
- Consistent operational quality

rather than operational speed metrics alone.

The findings also highlight the competitive nature of India's food delivery ecosystem, where customer experience, pricing strategy, and demand generation collectively drive long-term business growth and profitability.

ANALYSIS BY: SHUBHAM KULKARNI

In []:

In []: